



OPEN Precise ablation of cholesteatoma using a 445-nm diode laser

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In cholesteatoma surgery, precise tissue resection is important, as the diseased tissue needs to be completely removed while preserving the ossicular chain to prevent recurrence and hearing loss. The ability to precisely ablate tissue using a pulsed 445nm diode laser operating in single-trigger mode was investigated. Single laser pulses with a duration of 100ms were applied with a power ranging from 0.5 to 4W via a multimode optical fiber. Fiber distance to tissue was varied from 0.5 to 5mm, and irradiation angles from 20° to 90°. Porcine ear cartilage with 500µm remaining perichondrium on top served to represent the human ossicle structure with cholesteatoma. Further ablation experiments were performed on human cholesteatoma and ossicle samples. Ablation craters were measured by optical coherence tomography (OCT), and tissue damage assessed with haematoxylin and eosin (H&E)-stained sections. The ablation depth reached up to 350µm when operating at 4W power output without causing damage to the underlying cartilage. Laser pulses with 1W successfully ablated cholesteatoma, but did not damage the ossicles. The results show that a 445nm diode laser pulses can be used to effectively remove cholesteatoma, potentially preserving ossicles and hearing ability.

Keywords Cholesteatoma, 445nm diode laser, Single-pulse ablation, Middle ear surgery, Ossicular preservation

Chronic middle ear infection (otitis media) is one of the most common ear diseases worldwide and is estimated to affect more than 20 million people¹. Approximately one quarter of the patients develop cholesteatoma over time – a special form of otitis media characterized by the ingrowth of keratinizing squamous epithelium into the middle ear^{2–4}.

Cholesteatomas often remain undetected for years, and initial symptoms such as tinnitus, purulent otorrhea, or increasing hearing loss usually only appear once the ossicular chain has already been damaged^{5,6}. Patients with cholesteatoma have to be treated with surgery⁷. Although the standard procedure for mastoidectomy has been shown to be effective, it can be associated with complications such as hearing loss, delayed wound healing, facial nerve paralysis, vertigo, loss of taste, and an increased risk of infection^{1,8,9}. Depending on the clinical situation, mastoidectomy can be performed with or without creation of a radical cavity. To illustrate the clinical situation, Fig. 1 shows examples of the delicate morphology of the stapes and cholesteatoma tissue.

The invasive growth of cholesteatoma and the associated destruction of bone structures in the middle ear is associated with inflammation and immunological processes and represent a considerable challenge in otology^{4,10,11}. Given the delicate anatomy of the middle ear, particular attention must be paid to the small dimensions of the structures that need to be preserved. The stapes typically measures ~ 2.5 – 4mm in height and ~ 1.9 – 3mm in width at the footplate, whereas the fine structures of the posterior and anterior crus measure between 0.3 and 0.8mm^{12–17}. The width of the incus is ~ 2.1 – 5mm^{12,13,18,19}. The head of the malleus has a width of approx. 1.1 – 2.5mm and at the neck it has a width of 0.5 – 1.9mm^{12,13,20}. This range (~ 300µm to 5mm) reflects both the fine structural thicknesses and the overall size of individual components within the middle ear. Surgeons need to perform their procedures with precise accuracy because the ossicular chain has fragile structures. The effective removal of diseased tissue using laser-based techniques shows good potential, as these methods enable controlled tissue removal with minimal damage to the surrounding healthy tissue and the delicate middle ear ossicles. Preservation of the ossicular chain has been shown to be associated with improved postoperative hearing outcomes, and laser-assisted techniques may facilitate this by enabling precise tissue removal^{21–23}.

As a result, minimally invasive procedures are becoming increasingly important. In general, laser technology offers promising possibilities in this context, as it enables the efficient and gentle removal of diseased tissue^{22,24,25}.

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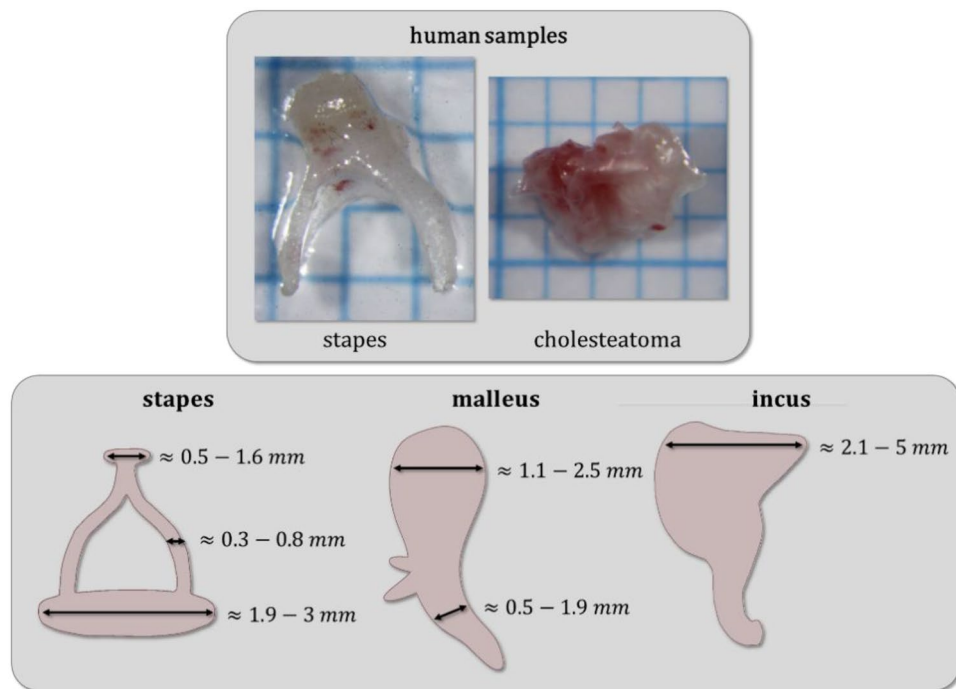


Fig. 1. Human stapes (left) and cholesteatoma specimen (right) on a millimeter scale grid and schematic outlines of human auditory ossicles: malleus, incus, and stapes. With representative width/thickness ranges at marked points.

Lasers used in ENT surgery range in wavelength from the infrared (CO_2 -laser: $10.6\mu\text{m}$; Er:YAG laser: $2.94\mu\text{m}$) to the visible region (Argon: 514nm , Nd:YAG laser frequency doubled with a KTP crystal: 532nm)^{22,26}. With an absorption coefficient of water of $\geq 80\text{mm}^{-1}$, the CO_2 laser radiation rapidly vaporizes tissue and coagulates small blood vessels simultaneously in a very thin zone. However, for guiding $10.6\mu\text{m}$ light for endoscopic surgeries, hollow-core fibers have to be used that are less flexible and more expensive than standard fibers²⁶. Also, for hemostasis in vascular tissue, a slightly greater penetration depth and high absorption specifically in hemoglobin is beneficial. In cholesteatoma surgery, the KTP laser with an absorption coefficient of 0.04mm^{-1} in water and $21 - 23\text{mm}^{-1}$ in blood ($150\text{g HbO}_2/\text{Hb}$ per litre) is preferred so far²⁶⁻²⁸. Since some years, diode lasers with a wavelength of 445nm are available with a higher absorption coefficient in hemoglobin and therefore an excellent ability of coagulation with a limited depth of penetration during surgical incisions ($\mu_A(\text{blood}) = 43 - 195\text{mm}^{-1}$)^{24,28,29}. Compared to other photoangiolytic lasers like the KTP laser (and pulsed dye lasers), the 445nm blue laser is, in addition to the increased absorption in hemoglobin, utilizing an enormous technical simplification by pumping the laser with a semiconductor (diode), thus avoiding complex technical constructions and maintenance services³⁰. Combined with its ability in noncontact cutting and availability in flexible fibers, it is ideal for endoscopic, single-handed surgeries, allowing the laser to reach and obliterate disease at angled anatomical sites³¹. The 445nm diode laser was shown to be versatile and effective in other subspecialties of otorhinolaryngology^{32,33} and Tam et al. showed in their retrospective cohort study that its use is safe in cholesteatoma surgery³¹. After partial microsurgical removal, the 445nm diode laser can be used in clinical practice to specifically eliminate any remaining cholesteatoma tissue.

The aim of this work is to investigate the use of a 445nm diode laser in single-pulse mode for the ablation of cholesteatoma tissue while preserving the ossicular structures. Pig ear cartilage was used as a tissue model to investigate ablation with single pulses of a 445nm diode laser, porcine tissue represents a well-established experimental model with comparable optical properties³⁴⁻³⁶. For example, the absorption coefficient of human cholesteatoma at 445nm (average value $\sim 3\text{cm}^{-1}$) lies near the range reported for soft tissues such as subcutaneous adipose tissue ($\mu_a \approx 1.8\text{cm}^{-1}$), and similar values have been reported for the reduced scattering coefficient (cholesteatoma: $\mu'_s \approx 4 - 7\text{mm}^{-1}$, subcutaneous adipose tissue: $\mu'_s \sim 1.4\text{mm}^{-1}$)^{37,54}. To simulate clinically realistic conditions, the samples were covered with a thin layer of blood, and single pulses, partly in overlapping sequences, were applied to replicate surgical resection paths. The pulse duration and power range were selected based on previous pulsed-laser studies and the concept of thermal relaxation time, aiming to limit heat diffusion and confine the effect to superficial tissue layers^{24,38,39}. The main objective is to determine parameter values which support both safe operation and reliable application results. The analysis of the influence of laser power, fiber distance, angle of incidence, and pulse overlap should show how these parameters determine the ablation behavior of the tissue. In addition, samples of human cholesteatoma and middle ear ossicles were used to test the transferability of the results to clinical conditions. In the long term, the findings may contribute to improving the precision of laser-assisted procedures, potentially reducing the risk of thermal or mechanical damage to the ossicles, and thus may supporting the preservation of hearing.

Results

Ablation of superficial tissue layers on cartilage model using single pulses

The ablation depth showed an increase in both groups in the power range of 0.5 – 4W for single 100ms-pulses (Fig. 2a). Each data point represents one measurement on a tissue sample, with the number of measurements varying due to differences in sample availability. This experiment involved analyzing both porcine cartilage and human cholesteatoma tissue to verify the comparability of the ablation depths. Ablation depths of $82 \pm 25\mu\text{m}$ were achieved in human tissue and $145 \pm 54\mu\text{m}$ in porcine tissue when operating at power levels between 0.5 – 1W. At an average power of 2W, the values in the porcine model were $247 \pm 85\mu\text{m}$, with no macroscopic damage to the underlying cartilage. At 4W, maximum ablation depths of $351 \pm 88\mu\text{m}$ (human) and $372 \pm 58\mu\text{m}$ (porcine) were achieved. Differences between human and porcine tissue were minor overall, confirming the porcine model as suitable. Linear fits (based on the steady state thermal ablation model, see discussion) result in a R^2 value of 0.75 for human cholesteatoma and 0.7 for porcine cartilage with residual tissue, supporting the use of a linear approximation within the investigated parameter range. The relatively large standard deviations can be attributed to the inherent variability of biological tissue, including differences in tissue thickness, composition, and blood content, as well as small variations in manual fiber positioning during the experiments.

The histological analysis supported the OCT measurement data. H&E staining revealed precise ablation of the superficial tissue layer (perichondrium), while the cartilage matrix remained intact throughout (Fig. 2b). Across all tested power settings, the underlying cartilage showed no structural alterations. The outer tissue layers were successfully removed without damaging the cartilage structure beneath.

Influence of varying fiber position on the ablation depth

In several fiber guide variations, the effects of fiber spacing and the angle of incidence (angle between the fiber and the tissue surface) on the ablation depth in cartilage with a residual layer were systematically investigated. The results demonstrated that both parameters affected the ablation depth, and the data confirm that precise tissue removal is achievable even with manual fiber guidance (Fig. 3).

The ablation depth decreases with increased distance between the fiber and the tissue surface. The average ablation depth reached $260 \pm 58\mu\text{m}$ at 0.5mm distance but it decreased to $215 \pm 55\mu\text{m}$ at 1mm and then reached $170 \pm 40\mu\text{m}$ at 2mm (2W, 100ms pulse duration, Fig. 4a). OCT images confirm that the ablation remains limited to the superficial residual layer, while the underlying cartilage remains structurally intact.

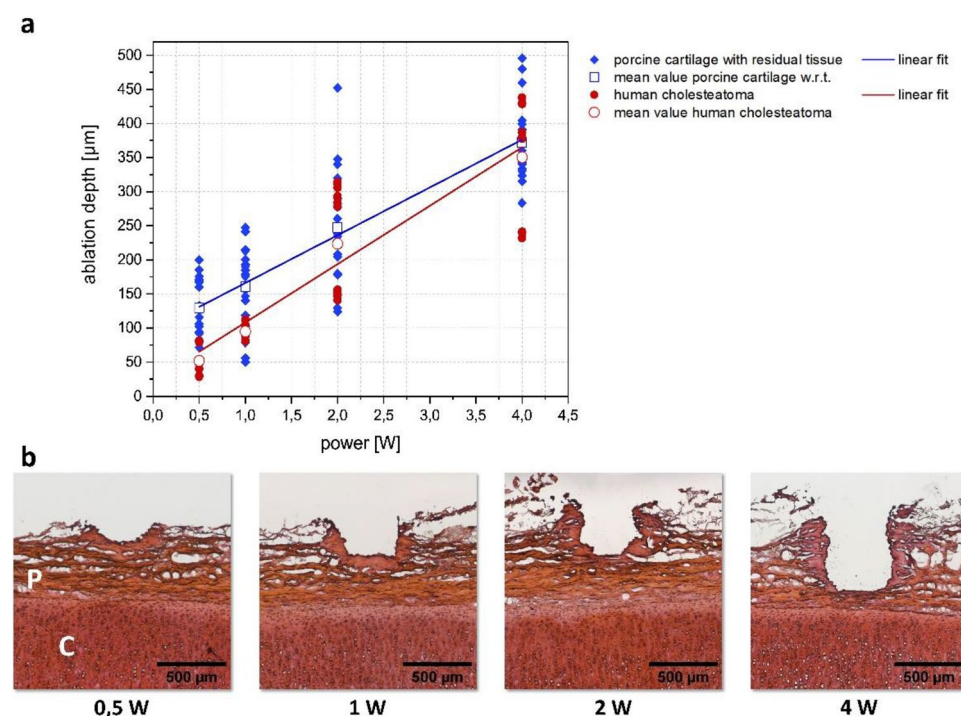


Fig. 2. Ablation depth and cartilage integrity after 445nm single-pulse laser ablation (100ms pulse duration, 1mm distance). **a**) Ablation depths in human cholesteatoma (red) and porcine cartilage (blue) measured by OCT. Nearly linear increase in ablation depth with rising laser power from $82 \pm 25\mu\text{m}$ (human) and $145 \pm 54\mu\text{m}$ (porcine) at 0.5 – 1W to $351 \pm 88\mu\text{m}$ and $372 \pm 58\mu\text{m}$ at 4W. **b**) H&E staining of porcine cartilage with residual tissue after laser ablation. The superficial layer was ablated depending on the power used, while the underlying cartilage matrix remained intact. P = perichondrium, C = cartilage. Scale bar: $500\mu\text{m}$.

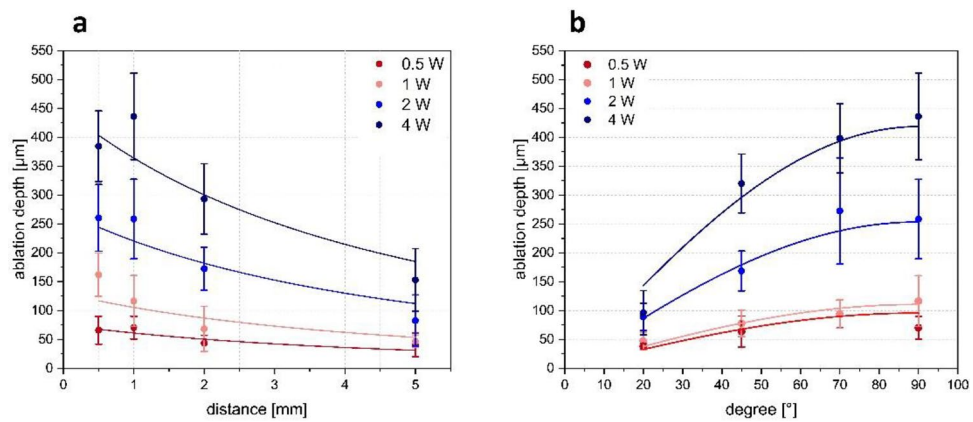


Fig. 3. Ablation depth as a function of distance (**a**) and beam angle (**b**) at different laser powers. Greater distance reduces ablation depth, while greater beam angle increases it, with the effects being more pronounced at higher powers. Values shown as mean $\pm$ standard deviation.

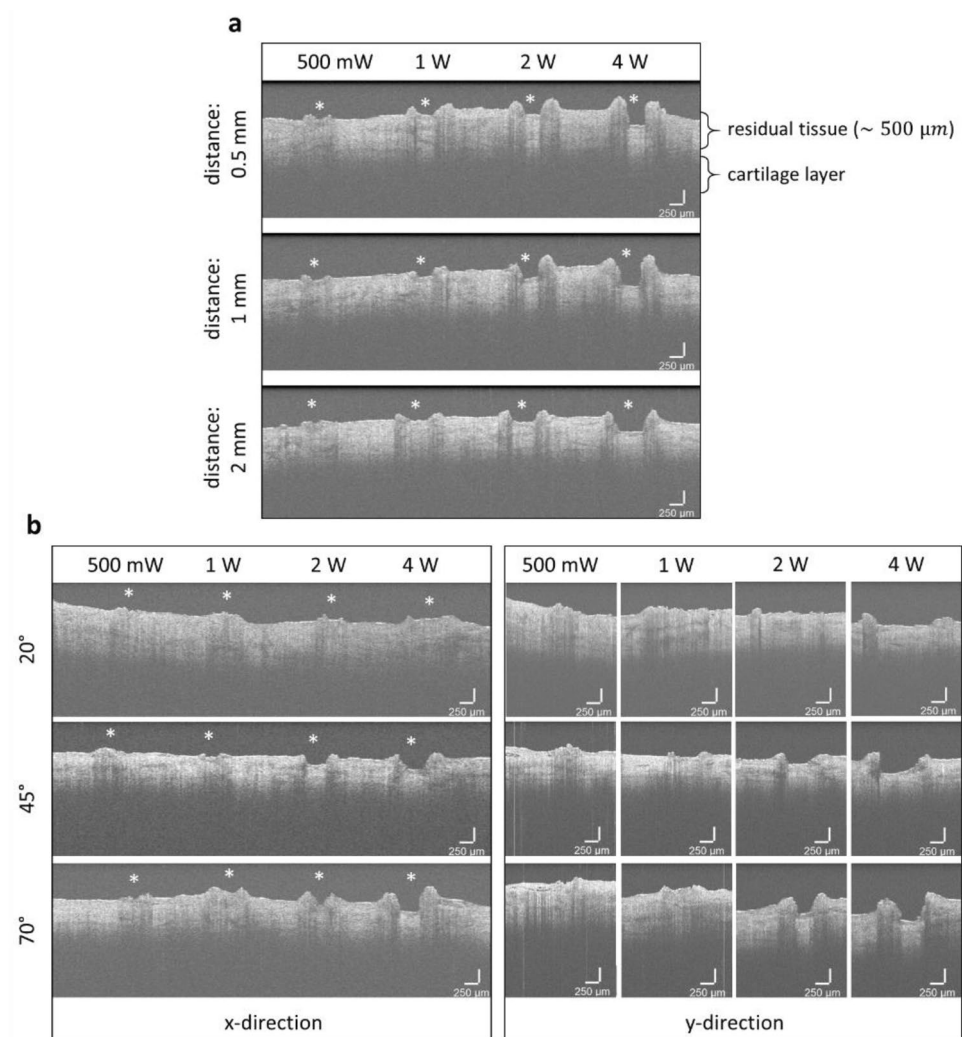


Fig. 4. OCT images of cartilage with residual layer after laser ablation at various distances (**a**) and angles (**b**) (445nm, 100ms pulse duration). Stars (*) mark ablated areas. Ablation is limited to the superficial layer; the underlying cartilage remains intact. Scale bar: 250 μm .

The fit curves shown in Fig. 3a were adjusted using the equation $y = b \cdot \left(\frac{r_f}{r(z)} \right)^2$, with r_f being the fiber radius and $r(z)$ the spot radius on the tissue for distance z (coefficient of determination $R^2 = 0.64/0.80/0.89/0.89$ for $0.5W/1W/2W/4W$). With this equation the ablation depths follows the quadratic decrease in fluence corresponding to the increase in area (assumption: steady-state model of thermal ablation, see discussion).

The angle of incidence influences both the shape and depth of the ablation crater. Flat angles produced wide, flat, and elliptical ablation zones, while a vertical incidence (90°) enabled the deepest ablation. The average ablation depth increased continuously with increasing angle (Fig. 4b). At 45° , depths of $270 \pm 90 \mu\text{m}$ were achieved, while at 70° , depths of $370 \pm 134 \mu\text{m}$ were already achieved. In the maximum configuration ($4W, 90^\circ$), the values were up to $435 \mu\text{m}$. This increase can be attributed to the smaller irradiated area and the resulting higher energy density at a steeper angle of incidence. The OCT images in Fig. 4 illustrate this angle-dependent increase in ablation depth.

This increase can be attributed to the smaller irradiated area and the resulting higher energy density at a steeper angle of incidence: The spot size on the tissue is proportional to $\sin(\beta)$ (Fig. 8 for explanatory drawing). Accordingly, the fit curves shown in Fig. 3b were adjusted using the equation $y = d_{fit} \cdot \sin(\beta)$ ($R^2 = 0.67/0.88/0.98/0.94$ for $0.5W/1W/2W/4W$).

Under manual fiber guidance, small variations in fiber distance of up to 1mm and angle variation of $5 - 10^\circ$ can be assumed based on the experimental setup and handling conditions, resulting in differences in ablation depth of approximately $50 - 100 \mu\text{m}$ for each parameter varying separately or up to $140 \mu\text{m}$ combining distance and angle variations (assumption: quadratic error propagation).

Ablation with overlapping pulses

Overlapping single pulses with a power of $2W$ and a pulse duration of 100ms were applied to cartilage tissue with a residual layer to simulate clinical common situation where tissue removal requires repeated laser application rather than a single pulse. The spot diameter was approximately $635 \mu\text{m}$, as calculated from the fiber geometry and divergence angle (Fig. 8) and the distances between the spots were 0.3 and 0.4mm . This overlap created a continuous ablation area. The OCT analyses showed uniform ablation patterns with depths ranging from 260 to $370 \mu\text{m}$, depending on the remaining tissue thickness (Fig. 5). Particularly with a thinner residual tissue layer on top, uniform and controlled ablation was achieved. The OCT data were used for quantitative evaluation of the ablation depth, while the cross-sectional images in Fig. 5 illustrate the ablation morphology. Across all examined samples, no structural damage to the underlying cartilage was observed. Given that the thickness of the residual tissue layer ranged between approximately 420 and $560 \mu\text{m}$, the measured ablation depths were always restricted to the soft tissue layer above the cartilage.

Laser ablation of human ossicles and cholesteatoma tissue

In order to validate the results from porcine tissue models, comparative studies were performed on human samples (445nm , 100ms pulse duration, $0.5 - 4W$, results depicted in Fig. 2a). Both cholesteatoma tissue and malleus and incus bones were used. Porcine blood was additionally applied to the cholesteatoma sample, but not to the ossicle samples.

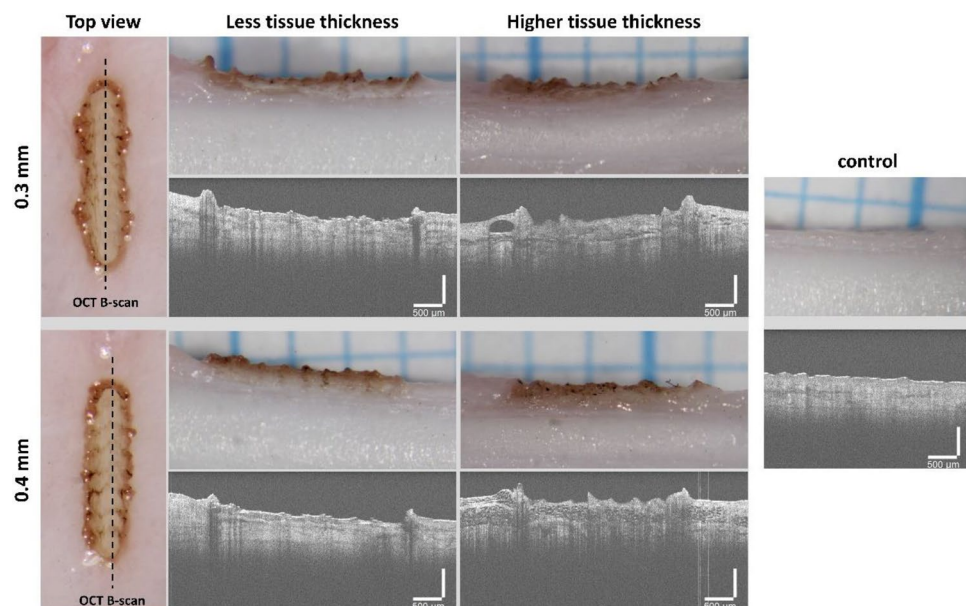


Fig. 5. OCT image of overlapping laser ablations with spot distances of 0.3mm (top) and 0.4mm (bottom). Top-view images together with representative macro- and OCT cross-sectional views of samples with different residual tissue thickness. The images demonstrate that superficial tissue residues on the cartilage surface could be removed without detectable structural changes in the cartilage cross-section. Scale bar: $500 \mu\text{m}$.

The OCT images showed clearly visible morphological differences between the tissues. As the laser power increased, the crater size in the porcine tissue increased continuously and was predominantly round to angular in shape, with superficial bulges indicating the remaining residual tissue. In contrast, the human ossicles showed only pointed craters or no ablative changes at all (Fig. 6a).

At a power of 1W, no ablation could be detected in the cartilage tissue of the human ossicles. At a power of 4W, signs of damage to the ossicles were visible ($\sim 250\mu\text{m}$). At a power of 2W, the ablation depths were clearly smaller ($\sim 90\mu\text{m}$). In contrast, cholesteatoma tissue showed clear and reproducible ablation effects at 2W and a depth of up to $315\mu\text{m}$. Overall, the human ossicles exhibited significantly reduced ablation behavior (Fig. 6b). At 4W, the ablation depth was only about 40–50% of the values measured in porcine cartilage samples and in cholesteatoma. Since there was no residual soft tissue or only small blood vessels present on the human ossicles, this lower ablation depth results from the lower absorption of mineralized bone compared to soft tissue or blood-covered tissue. This reflects the clinical situation, in which cholesteatoma matrix is often present on the periosteum of the ossicles, which must be carefully removed to prevent recurrence while preserving hearing.

Discussion

As already mentioned in the introduction, ENT surgeons can choose from a range of lasers with different wavelengths and pulse characteristics. The 445nm diode laser, due to the high absorption in hemoglobin and melanin, is expected to have good photoangiolytic properties and reduce intraoperative bleeding and scarring⁴⁰. While differentiation from CO₂ lasers is relatively clear due to the significant difference in absorption in water and hemoglobin, the question arises as to how the 445nm wavelength compares to “green” lasers such as the KTP laser with 532nm.

Several studies report that the 445nm laser has similar, but superior properties to KTP laser, including a better hemostatic effect, minimal water absorption hence minimizing injuries to the inner ear, and higher tissue absorption rate, which allows operating at lower power hence less thermal injuries³¹. Actually, there are studies showing an advantage of the 445nm laser compared to the KTP laser in laryngology⁴⁰. Lin et al. e.g. found that the ‘blue laser’ resulted in less vocal fold scarring compared to the KTP laser in normal rat vocal folds⁴¹. Balouch et al. conducted a study on 31 subjects diagnosed with vascular lesions of the vocal folds, in which the blue laser was used in pulsed mode at low power densities (3–4W). They found that it was less effective in preventing recurrence of vascular lesions on the vocal folds compared to the KTP laser. However, postoperative edema and haemorrhage were less severe in patients treated with the blue laser, indicating reduced thermal trauma^{40,42}. In contrast, Hanke et al. performed an experimental study with gingiva tissue with diode lasers ranging from 400 to 1500nm and found a higher cutting efficiency for 445nm compared to the near-infrared range (810nm, 980nm, 1064nm), but not to 514nm and 532nm²⁴. For cholesteatoma surgery, another aspect of 445 nm radiation might be beneficial: blue light has been utilized in other areas of medicine for bacterial decontamination^{43,44}. However, Tam et al. found no reduction in residual rate in a study including 120 cholesteatoma cases ($n = 47$ surgeries using the laser)³¹. Further studies are ongoing.

Based on these considerations, this research demonstrates the high potential of the 445nm diode laser for precise ablation in the middle ear region. With single laser pulses, small tissue volumes with a thickness $< 500\mu\text{m}$ can be removed. Cholesteatoma tissue with a thin blood layer was effectively ablated even at low power levels, whereas the compact structures of the ossicles showed a higher ablation threshold (Fig. 6). These differences in ablation behavior confirm the possibility of targeted removal of pathological areas while preserving functionally relevant structures.

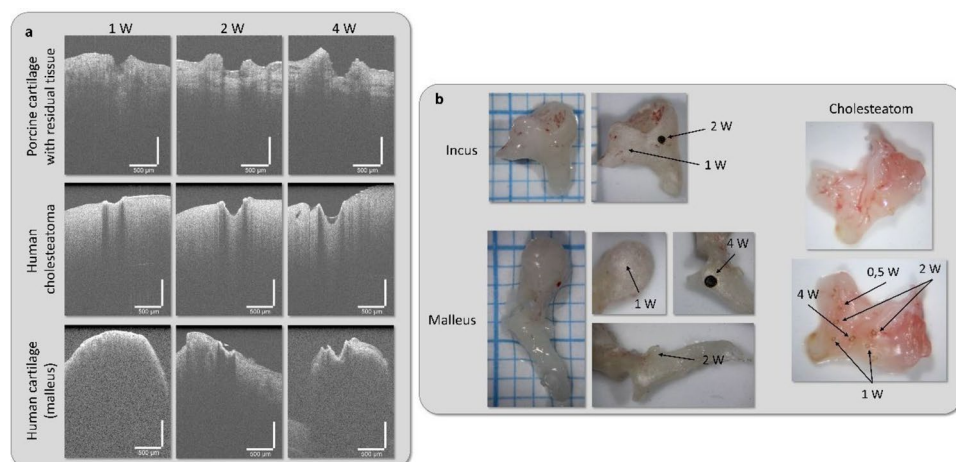


Fig. 6. **a**) Representative OCT images after laser ablation at 445nm on porcine cartilage samples with residual tissue (top), human cholesteatoma tissue (middle), and human ossicles (bottom). Scale bar: 500μm. **b**) Ex vivo samples of human ossicles (incus, malleus) and cholesteatoma tissue after laser treatment with different power levels. The lesions caused by the respective laser power (0.5–4W) are marked.

The ablation depth can be controlled by various parameters, such as power, fiber spacing, and beam angle. This has also been confirmed in other studies. Braun et al. demonstrated that a wavelength of 445nm in soft tissue enables high cutting performance with minimal thermal damage to the surrounding areas³⁹. With a steady-state ablation model (see e.g.³⁸), neglecting heat diffusion or plume shielding, the ablation depth is given by

$$d = \frac{1}{\mu_a} \ln \left(\frac{F}{F_{th}} \right)$$

with μ_a being the effective absorption coefficient, F the fluence and F_{th} the threshold fluence. Using a Taylor approximation of the logarithm, the depth thus follows a linear relationship with power when this model is valid. Hanke et al. showed a nearly linear relationship between power and incision depth for different diode wavelengths ($0.5 - 4\text{W}$, continuous wave (cw) operation), which corresponds to our single-pulse measurements²⁴. The depth values we measured with single 100ms pulses are also consistent with a linear dependence – the R^2 value for the fit on the human cholesteatoma samples is 0.75 and 0.7 for porcine tissue. The high intersection point of the adjustment curve for the pig data with the y-axis is noteworthy, as an intersection point $y \leq 0$ is expected for a threshold process such as thermal ablation. This will be investigated further. However, it should be noted that there is considerable variation in the values. In human cholesteatoma samples, this is already due to the considerable variation in tissue type and blood content in the samples. The homogeneity is significantly higher in animal tissue samples. Here, inaccuracies in fiber positioning, possible contamination of the fiber during pulse emission, and varying onset of ablation of weakly absorbing tissue (low blood content) contribute to the variance. Nevertheless, at a distance of 1mm and medium power levels, the values all stay within a range of a few hundred micrometers, without macroscopic damage to the cartilage.

Even at maximum power, the depths remained below the critical dimensions of delicate ear structures at around $350 - 370\mu\text{m}$ (Fig. 2a). This clearly indicates that a single pulse would not cut through the ossicles. These dimensions are surgically relevant, as the anatomical vulnerability points of the ossicles are of a similar order of size. The total height of the stapes is typically $\sim 2.5 - 4\text{mm}$, with the crus thickness at its narrowest point being $\sim 0.3 - 0.8\text{mm}$ ^{12–17}. The widths of the malleus and incus are in the millimeter range^{12,13,18–20}. Ablation depths below $300\mu\text{m}$ are therefore surgically advantageous, as they allow very precise tissue removal with less than 0.5mm ablation depth per pulse and ensure that a single pulse does not completely transect the ossicles. In all cases, histological examination of the underlying cartilage revealed no structural damage. The procedure thus demonstrated its effectiveness as a method for removing cholesteatoma tissue while preserving the ossicular chain, which could be confirmed in a clinical study using the 445nm wavelength laser⁴⁵.

The distance between the fiber and the tissue has a strong influence on ablation. As the distance increases, the spot diameter increases, the energy density decreases, and the ablation depth decreases³⁰. The ablation width in porcine cartilage with remaining tissue reached $450\mu\text{m}$ at 0.5mm distance but expanded to $920\mu\text{m}$ at 5mm distance when using 4W power. At low power ($0.5 - 1\text{W}$), the ablation width even decreased with increasing distance. The results confirm a quadratic correlation between ablation depth and distance³⁰. In clinical practice, distances of $0.5 - 2\text{mm}$ are common, so the aim is to achieve precise ablation with 100ms single pulses. Under these realistic conditions during manual fiber guidance, small deviations in distance and angle are inevitable and lead to slight variations in the measured ablation depth, typically in the range of $50 - 100\mu\text{m}$.

The small differences in ablation depth occur because of minimal variations in local energy delivery but remain within an acceptable range. The laser power requires precise optimization based on tissue structure because both uneven surfaces and manual guidance methods impact the achieved results. Nevertheless, this indicates that controlled tissue removal can still be achieved even under manual handling. In addition, the complex geometry and curved surfaces of middle ear structures such as the ossicles make it difficult to maintain a constant irradiation angle, which may further influence the local fluence and thus the ablation depth. The angle of incidence also plays a decisive role. If the fiber is held at an angle to the surface, an elliptical laser spot is created. Flatter angles result in a larger irradiated area and thus lower energy density⁴⁶. Our data show that at 45° degree and 4W , the deepest ablation was at the point of the elliptical crater closest to the fiber end, where the intensity of the Gaussian beam is highest⁴⁷. It can be assumed that angles between 45° and 70° are used in clinical practice. Measurements of the ablation depth in porcine cartilage with residual tissue at 2W ranged between 170 and $270\mu\text{m}$. Irregular surfaces and manual guidance can lead to variations, although these remained within the safe range.

Letting the laser spots overlap $26 - 42\%$, uniform resection paths could be created without enlarging the damage zones in the peripheral area. The OCT data showed that ablation depth stayed constant between 260 to $370\mu\text{m}$ in all examined residual tissue thicknesses. In addition, the ablation depth can be controlled via the pulse energy. Thicker tissue layers can be removed step by step, which is an important aspect in avoiding complications. The 445nm laser therefore can be used for tissue resections without damaging ossicular structures.

In this study, OCT was primarily used to quantify the ablation depth. However, OCT has increasingly been investigated as an imaging modality for middle ear structures. Several studies have shown that OCT can be used to visualize middle ear structures and may help to distinguish cholesteatoma from surrounding tissue^{48–51}. These findings suggest that OCT may not only serve as a quantitative tool but also has potential for intraoperative assessment of middle ear pathology.

The amount of control over the depth of ablation does not necessarily correlate with clinical “safe” versus “unsafe” threshold values. Available literature provides mainly qualitative recommendations rather than clearly defined limits. For example, Yau et al. state in their paper on endoscopic cholesteatoma surgery using a 532nm laser that, in their experience, the laser should not be focused directly on the bone for an extended period, as this can result in bone injury. They also note that lasering should be stopped immediately once charring of the bone is visible⁵². However, carbonization already indicates at least superficial damage to the bone. The authors

also mention that a perilymphatic fistula may occur when the stapes footplate is lasered. They suggest first removing gross disease with instruments, followed by the application of a defocused laser to any residual tissue. The laser was most frequently used to ablate cholesteatoma from the stapes superstructure. Tam et al. note that, in cholesteatoma surgeries using the 445nm diode laser, they took care to avoid applying the laser to vulnerable structures, including the facial nerve, the oval and round windows, and the dehiscence of the semicircular canals³¹. In addition, there are many tissue characteristics that can vary from patient to patient, and can significantly affect the surgical treatment (e.g., thickness of the bony specimen, inflammation, vascularization).

In CW mode, heat input accumulates due to continuous energy delivery, whereas in single-pulse mode, the energy is delivered for a short time and the tissue is able to cool down in between pulses. If the pulse duration is within the range of the thermal relaxation time, the effect is limited to the superficial layers⁵³. Studies on gingiva showed significantly broader and deeper thermal influence zones in CW mode, which further supports the use of single-pulse mode for working close to bone²⁴.

Blood absorbs strongly in the range of 445nm. Wisotzky et al. showed that the absorption properties of cholesteatoma change significantly when blood is applied, since a characteristic hemoglobin peak occurs between 430nm and 555nm³⁷. In our experiments, porcine blood was applied to all samples except the human ossicles. A thin blood layer on the tissue surface can enhance energy uptake at 445nm due to these hemoglobin peaks, supporting effective soft-tissue ablation²⁵. In our experiments, porcine blood was applied to all samples except the human ossicles, allowing a direct comparison between blood-covered and non-covered tissue. The results demonstrate the ability to reliably estimate the ablation properties at this wavelength for both the tissue and the surrounding structures. Human cholesteatoma could already be removed at 1W, a power at which the ossicles showed no alterations. This can be explained by structural differences, as mineralized bone has a higher ablation threshold, and a lower attenuation, resulting in the laser energy being deposited in a larger volume. Nevertheless, some superficial ablations were also observed at 2 and 4W.

Regarding our results, the recommended clinical approach to support optimal ablation is to start with vascularized tissue or apply a thin layer of blood to the perimatrix of the cholesteatoma. The 445nm diode laser enables precise ablation in single-pulse mode by selecting the optimal power levels, distances and irradiation angles. It is a promising instrument for the controlled ablation of cholesteatoma tissue while protecting sensitive middle ear structures.

Conclusion

Single 445nm diode laser pulses enable precise ablation of cholesteatoma tissue while protecting the ossicular structures. The procedure therefore offers great potential for clinical use as a hearing-preserving supplement in cholesteatoma surgery.

Materials and methods

Preparation and characterization of porcine cartilage models

Cartilage samples from the external ear of pigs (slaughterhouse material) were used for the experimental investigations. The tissue was obtained commercially by the Medical Laser Center Lübeck from a slaughterhouse licensed according to European Union regulations (EC No. 853/2004 and EU 2017/625). The samples were transported to the laboratory in a cooled container and stored in a refrigerator at 4°C until further processing for a maximum of 3 h, with all experiments performed on the same day to ensure comparable tissue conditions.

After cleaning the auricle thoroughly under running water, the remaining auricle was cut into strips approximately 1cm wide, avoiding all cartilage ridges. The superficial skin and fat layers were then removed manually with a scalpel (Swann-Morton, UK) and dissecting scissors (Hammacher, Germany). A thin layer of residual soft tissue remained on the cartilage surface. This served as a reference tissue model for the cholesteatoma that can form over the cartilage. The exposed cartilage tissue was removed with tweezers (Fuhrmann, Germany) and cut into square samples with an edge length of approximately 1cm. The finished samples were transferred to sterile 6-well plates (Greiner Bio-One, Austria) and stored there in Ringer's solution until the experiment was carried out.

Optical coherence tomography (OCT) was used to determine the thickness of the remaining soft tissue layer on top of the cartilage (Telesto 1060nm with the OCT-LSM03-BB lens (Thorlabs, USA)). The thickness of the remaining tissue was between 420 – 560µm (Fig. 7). To increase absorption, a thin layer of porcine blood was applied on the sample surface before ablation (Fig. 7). This porcine blood was diluted with 3.8% sodium citrate in a ratio of 1 : 10 immediately after collection and frozen at –24°C. The blood underwent a controlled thawing process at 4°C throughout an entire night to achieve consistent conditions.

Human cartilage reference tissue

Human tissue samples were taken during clinical operations at the Department of Otolaryngology at the University Medical Center Schleswig-Holstein, Lübeck Campus. The samples were used after approval by the ethics committee (ethics vote: file number 2022–560). We confirm that all methods were performed in accordance with the relevant guidelines and regulations (Declaration of Helsinki). Immediately after collection, the tissue samples were stored in physiological saline solution and processed within two hours for ablation experiments.

Ablation experiments

Tissue ablation was performed using a 445nm diode laser (Fox IV, A.R.C.) with a 300µm core diameter optical fiber (HighFlex, A.R.C., numerical aperture 0.22). This laser can be operated in a mode in which individual pulses (in our tests: 100ms) can be triggered via the foot switch.

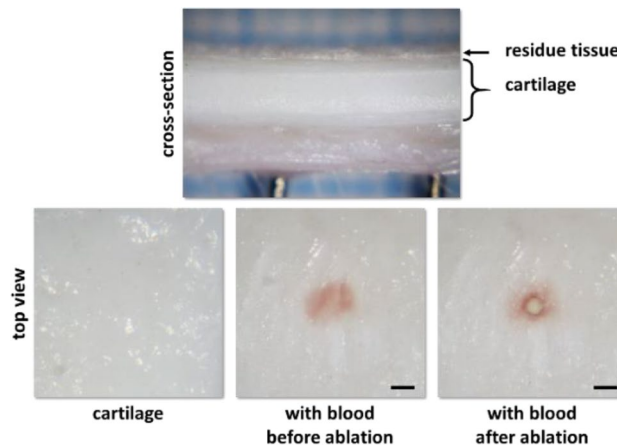


Fig. 7. Cross-section of cartilage and top views. The top shows a cross-section of cartilage, while the bottom shows intact cartilage, cartilage with blood spot, and cartilage after ablation. Scale bar: 1 mm.

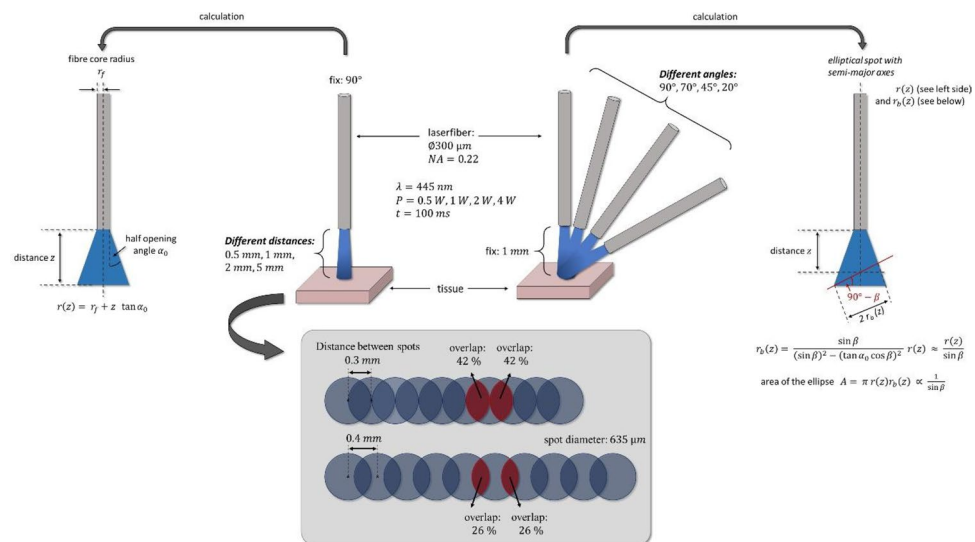


Fig. 8. Experimental setup for ablation. In single-pulse mode, either the distance between fiber tip and sample was varied (left) or the irradiation angle (right). Afterward the simulation of laser spot overlap. At 0.3 mm spacing, spots overlap by 42%, while at 0.4 mm spacing the overlap is reduced to 26%.

Before each ablation was performed, it was necessary to apply a small amount of pig blood to the samples. This also helped to maintain a moist surface and reduce potential drying effects during the measurements. Tissue samples from both porcine cartilage and human tissue were used. The blood was applied using a thin brush, and the surface of the sample was dried with blotting paper before application to ensure the smallest possible and optimal blood layer height²⁵. This enabled reproducible carbonization and simulated realistic surgical scenarios, which in turn allowed the volume removal and ablation depth to be measured. An image was then captured using a camera attachment (EOS 550D, Canon, Japan) on a microscope (Stemi SV8, Zeiss, Germany). The sample was then positioned on an adjustable sample table located below the fiber. The fiber was attached to a robotic arm (Rotrics DexArm, Rotrics, USA). The LM-10 thermopile sensor from Coherent USA measured power at the fiber end before each ablation test to confirm that test conditions remained constant.

In the following experiments, the fiber was aligned over this initiation point. In order to take into account different scenarios during an operation, the distances between the tissue sample and the fiber, as well as the angles of the fiber to the sample, were systematically varied. Measurements were taken at distances of 0.5 mm, 1 mm, 2 mm, and 5 mm from the sample, with the angle of the fiber to the tissue sample always being 90° (Fig. 8). Furthermore, the fiber was positioned at angles of 20°, 45°, 70°, and 90° to the sample, while the distance between the fiber and the sample was set to 1 mm for this series of measurements (Fig. 8). The pulse length of the single pulse was always 100 ms and was tested with different laser powers (500 mW, 1 W, 2 W, and 4 W, corresponding to the power emitted during the laser pulse).

Single pulses were used for the first experimental series, analogous to its use in surgical cholesteatoma resection. To simulate continuous tissue ablation, in another experiment series of 10 individual spots were applied on porcine cartilage samples (with a small amount of blood on top, as described above). The distance between the spot centers was 0.3 mm or 0.4 mm (Fig. 8).

In half of the samples, a tissue layer thickness of approximately 0.5 – 1 mm was left. A distinction is made between “less tissue thickness” (< 500 μm) and “higher tissue thickness” (> 500 μm). The applied laser power was 2 W with a pulse duration of 100 ms. The fiber spacing was kept constant at 1 mm. With a (previously measured) half opening angle α_0 of $\sim 9.5^\circ$ (Fig. 8), a spot width of 635 μm was calculated for this fiber distance. This resulted in overlapping intersection areas of approx. 42% (0.13 mm²) at a spot distance of 0.3 mm and areas of approx. 26% (0.08 mm²) at a distance of 0.4 mm.

Evaluation of ablation craters and tissue damage

Optical coherence tomography (OCT, see section “Preparation and characterization of porcine cartilage models”) was used to measure the depth and width of the laser ablation craters. The lateral resolution is 8 μm , the axial resolution 7.3 – 8.5 μm . Cross-sectional B-scans were analyzed to determine the distance between the crater edges and the deepest point. Each measurement was repeated three times to minimize error.

After OCT evaluation, the tissue samples were embedded in Tissue-Tek Cryomold molds (25 × 20 × 5 mm) using Tissue Freezing Medium (Leica), frozen at -80°C , and then cut into 20 μm thick sections using a 2800 Frigocut cryostat (Reichert-Jung). Sections were taken at different positions along the laser spot and stained with hematoxylin–eosin (H&E) according to a modified Carl Roth Rapid Staining Kit protocol.

Stained sections were examined under an inverted microscope (Nikon) at 4x and 10x magnification. Both overall images and detailed views of the ablated areas were analyzed, and the open-source software Fiji was used for quantitative evaluation of the damaged areas.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Author contributions

PE: methodology, data collection and analysis, visualisation, writing—original draft of the manuscript, editing AMD: data collection and analysis BL: methodology, data collection and analysis, writing ZP: drafting and submission of the ethics application DL: drafting and submission of the ethics application, patient recruitment RR: data analysis, writing, supervision KLB: drafting and submission of the ethics application, patient recruitment, supervision AL: drafting and submission of the ethics application, patient recruitment, supervision, project coordination RB: funding acquisition, supervision, resource and project coordination All author reviewed the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Approval of local ethics committee

Evaluation of human tissue was approved by ethics committee of University of Lübeck, Campus Lübeck, Germany, No.: 2022–560.

Additional information

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